CROSS-PILLAR CONSTRAINT INTERACTION & DEPENDENCY PATHWAYS

How Structural Deficits in One Pillar Propagate Downstream to Constrain National AI Capability

Pathway A: Spatial Clustering & Grid Lock-In **[Pillar 4 (Subsea Cables)] ---> [Pillar 3 (Data Centres)] ---> [Pillar 2 (Power Grid)]**

Subsea cable landings in Mumbai (17+ systems) attract 52.4% of DC build-out -> Creates 892 MW local demand shock -> Exceeds 220kV substation capacity -> Triggers 18--36 month interconnection queues -> Strands deployed compute silicon.

Pathway B: Tropical Thermodynamics & Water Competition **[Pillar 3 (Physical Resources)] ---> [Pillar 2 (Energy)] ---> [Pillar 5 (Social License)]**

Monsoon ambient heat (>42 deg C) spikes PUE to 1.55 -> Conventional wet cooling consumes 11,250 ML/yr at 500k scale -> Direct competition with municipal drinking water in Mumbai & Chennai -> Forces transition to D2C liquid cooling.

Pathway C: Semiconductor Chokepoint & Fiscal Reversion **[Pillar 1 (Semiconductors)] ---> [Pillar 5 (Governance)] ---> [Pillar 1 (Packaging Hub)]**

Pursuing leading-edge sub-3nm fab requires >$30B capex (7--10 yr delay) -> Cascade proves wafer starts are only 1.6% of TSMC -> CoWoS packaging is 7.41% global chokepoint -> Refocuses policy toward OSAT/ATMP (Sanand/Morigaon).

Pathway D: Linguistic Inefficiency & Compute Multiplier **[Pillar 4 (Indic Data)] ---> [Pillar 1 (GPU Demand)] ---> [Pillar 2 (Grid Draw)]**

Foreign LLM tokenizers allocate 4--8 tokens/word for Hindi/Tamil vs 1--2 for English -> 3--6x inference cost penalty -> Multiplies required compute stock -> Drives higher power grid demand -> Accelerates domestic model priority (Sarvam/BharatGen).